

WHAT STUDENTS DO

Drag members between grid joints and drive a vehicle across. Costing follows a fabricator's bill: **steel by weight** (9.4 kg/m at \$4.80/kg ≈ \$45/m) plus **\$180 per joint** for the gusset, bolts and labour. So the **cheapest bridge that survives wins**, golf-style against par. Members are coloured live — **red = tension**, **blue = compression**, **brighter = nearer breaking** — so the load path is visible before anything fails. Offline in any browser: open [index.html](#) (Chromebook-friendly, mouse or touch). Underneath is a real **direct stiffness** solver: element stiffnesses assembled into a global **K**, supports as boundary conditions, **K·u = F** solved for joint displacements, member forces from the strains. Overloaded members are deleted and the structure *re-solved*, so failures cascade like real ones.

30-MINUTE SESSION PLAN

- 0–4** Project it (⚙️ → **Projector mode**). On Level 1 build *only* the flat roadway and press **Load test**. It folds. Name what they just saw: joint, member, tension, compression, support. Add a post and two struts; run it again.
- 4–10** Students play **Level 1** in pairs with **X-ray** on. Have them use the **Joint** tool: tap a joint, see every force meeting there and their sum. It is always zero. That panel is the method of joints.
- 10–18** **Level 2 — the lesson**. It opens with a rectangular frame already built. Test it first: it folds into a parallelogram and the debrief reports a **mechanism**, naming the joints that ran away. Take predictions, then let them add diagonals. Par forces a second pass: which members do nothing?
- 18–25** **Level 3 — Heavy Haul**. Four times the load. Point them at the debrief's "**buckles at**" column to discover that compression rating depends on *length* and tension rating does not.
- 25–30** Project a student's debrief. Load Pratt, Howe and Warren from the **Gallery** at equal depth and compare. Post the leaderboard. Levels 4–6 make good homework.

WHY CONNECTIONS ARE PRICED

A flat price per metre makes extra panels free, which is false and hides a real design driver. Here connections are roughly **45%** of a typical bill, so panel count is a genuine trade-off: bracing a panel you already have costs steel only, while adding a panel costs a joint as well. That is exactly why real trusses use as few panels as they can. The debrief splits the bill into steel and connections.

AXIAL MEMBERS

Every joint is a pin, so it carries no bending moment. A straight member pinned at both ends is therefore a **two-force member**: its only possible force is along its own axis — a pull (**tension**) or a push (**compression**). One number describes it completely. Each joint satisfies **ΣF_x = ΣF_y = 0**; the Joint tool draws exactly that and prints the residual.

TENSION VS. COMPRESSION

Same steel section throughout (E = 200 GPa, A = 12 cm²). Pulled, it yields at T_{max} = σ_yA = **300 kN at any length**. Pushed, it does not crush — it **buckles sideways**, at

$$P_{Cr} = \pi^2 EI / L^2 \approx 720 \text{ kN}\cdot\text{m}^2 / L^2$$

capped at 180 kN when stubby. Length is **squared**: double a strut, quarter its capacity.

LENGTH	1 m	2 m	3 m	4 m
Pulled	300 kN	300	300	300
Pushed	180 kN	180	80	45

Students can find the consequence themselves: flip a working design below the deck so the long members *hang* instead of *prop*. Same geometry, same cost, far lower stress.

WHY TRIANGLES

Three side lengths fix a triangle completely — its corners cannot move without stretching a side. Four do not fix a quadrilateral: it shears into a parallelogram with every side the same length as before. A member only resists a *change in its own length*, so a rectangular panel meets that shearing with **zero** resistance. In the solver this is arithmetic, not a special case: the shearing motion strains nothing, so K maps it to zero force and is **singular**. The game finds the rank deficiency, reads the null-space direction out of the elimination, and folds the bridge *along that exact direction* — the collapse students watch is the real mathematical failure mode. During the build phase it reports **how many independent folds** exist (the dimension of that null space) and animates each in turn. A null vector has zero strain only to *first order*,

so the drawn shape is projected back onto exact member lengths before display — what the class watches really does keep every member at its built length. Adding one well-placed member removes exactly one fold: count down to zero.

STATIC DETERMINACY, PLAINLY

- Count members **m**, reactions **r**, joints **j**; each joint gives 2 equations.
- m + r < 2j** — *mechanism*. Not enough members. It folds; strength cannot help.
 - m + r = 2j** — *determinate*. Equilibrium alone gives every force, and **every member is critical**: lose one and the rest is a mechanism.
 - m + r > 2j** — *indeterminate*. Spare members; load now splits by **stiffness**, not equilibrium alone. This is why the game solves K·u = F instead of walking joint to joint.
- The count is necessary but not sufficient — members in the wrong places leave a local mechanism. The solver checks the matrix, not the count.

Misconception 1 — “more members always makes it stronger.” Extra members change how load *shares*, and a stiffer path attracts *more* force, so an added member can overload its neighbour. And every member costs money: the debrief flags those that never passed 5% of their limit, and first attempts usually have several. **Try it:** add a member to a design that just made par and watch the peak move.

Misconception 2 — “the verticals do nothing, they’re spacers.” In a Pratt the verticals are in compression carrying each panel's shear to the top chord; in a Howe they are in tension. **Try it:** load a Pratt from the Gallery, turn on X-ray, erase one vertical — its panel becomes a rectangle and the truss becomes a mechanism. A member genuinely carrying nothing is drawn grey, and that is a different thing.

Misconception 3 — “tension and compression are just plus and minus.” The *same* 4 m member holds 300 kN pulled and 45 kN pushed. The sign is not bookkeeping; it selects a completely different failure mode. **Try it:** note a design's peak use, then rebuild it upside down under the deck.

DISCUSSION QUESTIONS

- The bare rectangle folded without any member changing length. Where did the load go, and why could no member resist it?
- Two designs, same cost and same joints, but one props the deck from above and one hangs it from below. Predict which is safer, then check. What in P_{cr} = π²EI/L² explains it?
- A Warren's diagonals alternate tension–compression all the way along. Take one bottom joint and use ΣF_y = 0 to show why its two diagonals must have opposite signs.
- Why is the bottom chord's force greatest at mid-span and near zero at the supports?
- One member snapped and the whole bridge came down. What does that say about its spare load paths? What would you add, and what would it cost?
- Level 5's pier drops par sharply. Halving a span does much more than halve the forces — why? Where do real bridges use this, and when can't they?
- Turn on **Self-weight** and retest a design that just made par. What happens, and why does it matter more as spans grow?

LEVELS, PAR & LOGISTICS

#	LEVEL	SPAN / VEHICLE	Par	LESSON
1	First Crossing	6 m · car 30 kN	\$1,050	A flat deck is a mechanism
2	The Long Gap	12 m · car 30 kN	\$2,800	Rectangles fold, triangles don't
3	Heavy Haul	10 m · truck 125 kN	\$2,800	Buckling; length matters
4	High Water	12 m · van 60 kN	\$3,100	Through truss / bowstring
5	Island Pier	16 m · truck 125 kN	\$3,600	Supports beat steel
6	Sandbox	18 m · your choice	—	Free build

- ⚙️ **Teacher mode:** projector large-UI toggle, leaderboard resets, and an option to **pick the test vehicle on any level** — for showing what happens when a bridge that carried a car meets a 12 tonne tipper. Demo runs stay off the leaderboard.
- Boards and saved designs live in each machine's browser storage; reset between periods.
- Keys: **1 2 3** build / erase / joint · **Ctrl+Z** undo · **Space** test · **X** X-ray · **F** force numbers.